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Four-wave mixing effect on high-power continuous-wave all-fiber lasers

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A four-wave mixing effect on high-power continuous-wave fiber lasers has been demonstrated theoretically and experimentally. Detailed theoretical description of phase matching is presented and we found that the phase matching condition is satisfied at the frequency shift of 5.16 THz. While the intensity in fiber core region is more than about 394 MW/cm^2, the four-wave mixing products of 1100 nm and 1060 nm were also observed in high-power all-fiber laser. The comparison shows that the experiment result is in good agreement with the simulation result. In addition, the beam quality deterioration for the laser is caused by the four-wave mixing effect and the mode instability. The M^2 factor measured at maximal intensity of 478 MW/cm^2 is 2.80.

Keywords: Fiber lasers and amplifiers; four-wave mixing; ytterbium-doped fiber; nonlinear effect.

1. Introduction

High-power fiber lasers have become attractive light sources for many applications, such as in industrial processes, scientific research, national defense and military.[1–6] However, with the promotion of output power, the optical transmission in fiber core region easily results in generation of nonlinear optical effects, including stimulated Brillouin scattering (SBS), stimulated Raman scattering (SRS) and four-wave mixing (FWM).[7–12] Compared with the former two, FWM is an important nonlinear effect in large-mode-area (LMA) fiber because it has a low threshold for satisfying the phase match in high output power.[13,14]

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Recently, FWM has been successfully used to implement wavelength converters, fiber-optic parametric amplifiers, single and entangled photon pair sources and so on.[15–19] With the increase in fiber data transmission capacity of multimode optical communication systems, nonlinear phenomena in multimode fibers, particularly FWM, have received renewed attention.[20–24] Moreover, FWM in single-mode fibers, zero dispersion fibers and photonic crystal fibers has been studied.[25–29] However, the studies of FWM effect on high-power continuous-wave (CW) fiber laser are even fewer.

In this paper, the FWM effect on high-power CW fiber laser with ytterbium-doped double-cladding fiber (YDCF) has been demonstrated theoretically and experimentally. We found that the phase matching condition was satisfied at the frequency shift of 5.16 THz, corresponding to the anti-Stokes wavelength of 1060 nm and the Stokes wavelength of 1100 nm. Using 20-m home-made YDCF in the amplifier stage, the FWM products were also observed when the laser power intensity in the fiber core area was more than 394 MW/cm^2.

2. Phase Matching in Optical Fiber

In the high-power fiber lasers and amplifiers, the strong FWM results from the phase matching between the fundamental mode and the high-order mode.[30–32] The normalized wave number, or V parameter, is used to define the mode propagation characteristics of an optical fiber and is a dimensionless number.[33] It is mathematically expressed as

$$V = \frac{2\pi a}{\lambda}\sqrt{n_c^2 - n^2}\,,$$

where a, λ, n_c and n are the core radius of the fiber, wavelength, refractive index of the core and refractive index of the clad, respectively. As the momentum conservation of four photons needs to be satisfied, which implies that the wave numbers of two new photons have to be of opposite sign. Here, we assume that the anti-Stokes light is the fundamental mode at a shorter wavelength with the frequency shift of $+\Delta f$. Moreover, we assume that the Stokes light is the same higher-order mode at a longer wavelength with the frequency shift of $-\Delta f$.[34] The phase matching $\Delta\beta$ is expressed as

$$\Delta\beta = \beta_{mn}\big|_{f=f_0-\Delta f} + \beta_{01}\big|_{f=f_0+\Delta f} - \beta_{mn}\big|_{f=f_0} - \beta_{01}\big|_{f=f_0}\,,$$

where the subscript m represents the mth-order Bessel function that corresponds to the cutoff condition for the mode and n represents the number of successive zeros in the corresponding Bessel function.[35] β_{mn} represents the propagation constant of LP$_{mn}$ mode, and f_0 is the frequency of all-fiber laser.

Here we assume a step index fiber with the core diameter of 30 μm and the refractive index difference of 1.6×10^{-3}, corresponding to the normalized wave number of about 6. Figure 1(a) shows the calculated phase mismatch for the fibers of $V = 6$. In this case, one pump light is the fundamental mode with a central

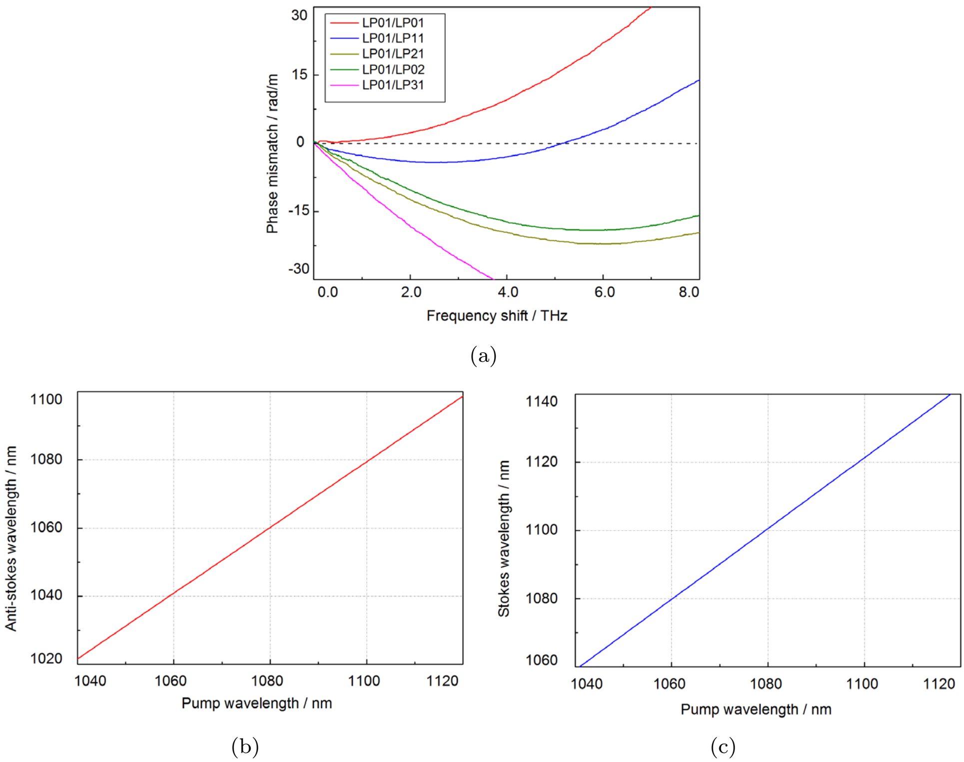

Fig. 1. (Color online) (a) The calculated propagation constant mismatches for the fiber of $V = 6$; (b) the anti-Stokes wavelength as a function of the pump wavelength; and (c) the Stokes wavelength as a function of the pump wavelength.

wavelength of $\lambda_0 = 1080$ nm, and the other pump light is one of the higher-order modes at λ_0. As shown in Fig. 1(a), the phase matching condition is satisfied at $\Delta f = 5.16$ THz when we choose LP_{11} mode as the Stokes light. The calculated anti-Stokes and Stokes wavelengths at different pump wavelengths are shown in Figs. 1(b) and 1(c), respectively. When pumped at 1080 nm, the anti-Stokes and Stokes wavelengths are 1160 nm and 1100 nm. Therefore, the FWM in the high-power fiber lasers and amplifiers occurred between the fundamental mode and the secondary mode.

3. Experiment and Method

As shown in Fig. 2, the experimental setup includes a master oscillator power amplifier (MOPA) configuration for high-power fiber laser. Six 600-W diode lasers (LDs) are applied as the pump sources which are coupled to the 30/600 YDCF via the $(6 + 1) \times 1$ multimode pump coupler. The fiber core diameter is 30 μm and the fiber cladding diameter is 600 μm. Two cladding power strippers were used to enhance the beam quality (BQ) by wiping off the residual pump power

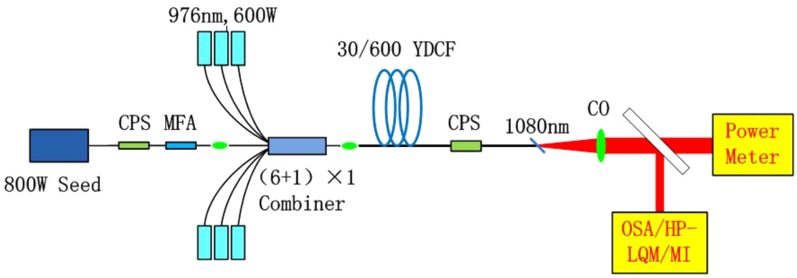

Fig. 2. MOPA configuration for high-power fiber laser.

and preventing the seed laser from the reflected power.[36–38] To avoid any end-face reflection, the home-made end-cap was spliced to deliver the output signal power. In addition, a mode field adapter (MFA) was adopted to decrease the splice loss between the seed laser and fiber combiner.[39,40] The spectrum was measured by optical spectrum analyzer (OSA, YOKOGAWA, AQ6370C) and the output power was recorded by a power meter (PM, OPHIR, 5 kW-ROHS). Also, the BQ under high-power condition was obtained via a high-power laser quality monitor (HP-LQM, PRIMES). To monitor the onset of mode instability (MI),[41] a photodetector (Thorlabs, PDA36A-EC) with a pinhole of 1.5-mm diameter was placed in the center of the collimated beam.

4. Results and Discussion

The laser output powers at different pump powers of fiber amplifier are shown in Fig. 3. In this experiment, the seed laser power of 800 W and the pump power of 3350 W were set up in the fiber amplifier. Then the maximum output power reached to 3379 W and the slope efficiency of fiber amplifier was 77.0%. Note that the laser output power linearly increases when increasing the pump power. In addition, the backward power was measured by a power meter and the maximum power of

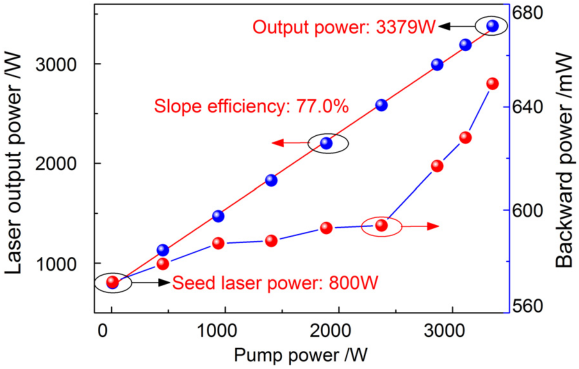

Fig. 3. (Color online) Measured output power versus the pump power of fiber amplifier.

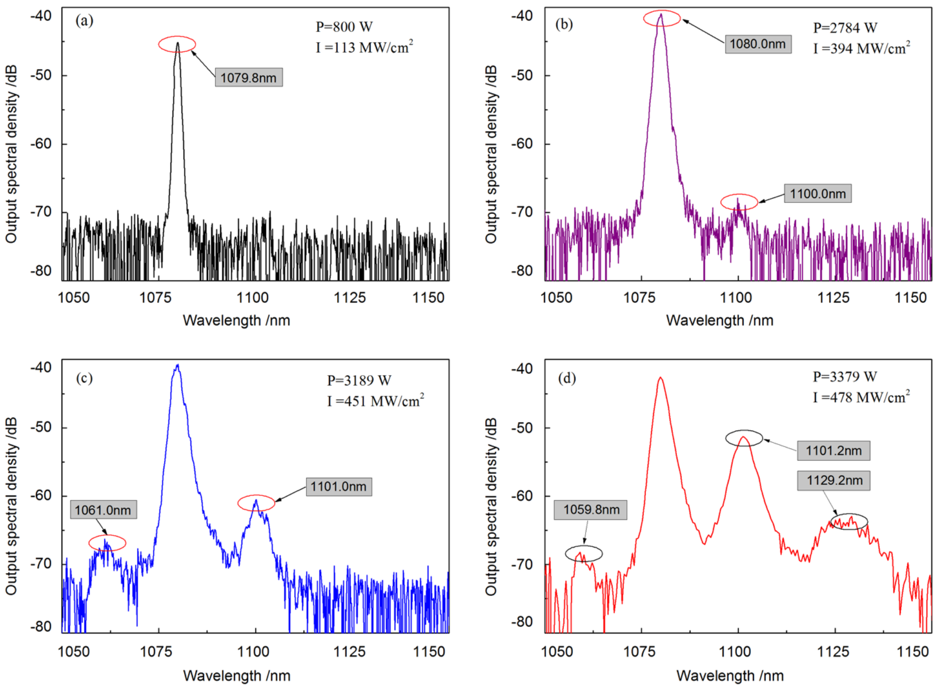

Fig. 4. (Color online) Measured spectra from fiber amplifier at different laser intensities: (a) 113 MW/cm^2, (b) 394 MW/cm^2, (c) 451 MW/cm^2 and (d) 478 MW/cm^2.

the backward light was tested as 650 mW in the experiment. The tested results show that the back reflections of Stokes and anti-Stokes are not amplified because they do not meet the phase matching condition between the forward propagating fundamental modes.

Figure 4 demonstrates the measured spectra of fiber laser at different laser intensities. The spectrum of the high-power fiber laser was measured using an OSA with 0.1-nm resolution. While the optical intensity in the fiber core area is lower than a critical value of about 100 MW/cm^2, the output spectrum contains only the pump wavelength $\lambda_0 = 1080$ nm [Fig. 4(a)]. At approximately 394 MW/cm^2, the Stokes line at the wavelength of 1100 nm is shown in Fig. 4(b), which indicates that the FWM phase-matching condition is satisfied with this fiber laser configuration when the laser output power is 2784 W. Moreover, the spectral broadening of high-power fiber laser arises from self- and cross-phase modulation effects. With a gradual increase in the laser intensity, the anti-Stokes line of 1060 nm appears at about 451 MW/cm^2, as shown in Fig. 4(c). In addition, the anti-Stokes line and Stokes line are amplified and broadened with the increase of the nonlinear interaction, but unequally. At the maximal output power, the power asymmetry between the two sidebands is about 20 dB in Fig. 4(d). From the output spectrum, we can find that a new wavelength peaking at 1129.2 nm appears and power begins to transfer into this wavelength partially. Based on a 440-cm^{-1} frequency shift in the silica optical

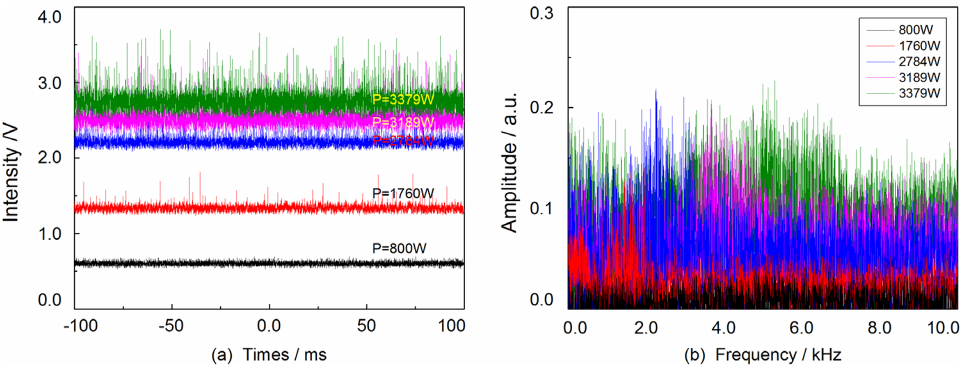

Fig. 5. (Color online) Fluctuation characteristics of the output beam: (a) time traces and (b) frequency distribution.

fiber, the Stokes wavelength (RSR) is about 1130 nm corresponding to the signal wavelength of 1080 nm. Therefore, this phenomenon can be attributed to the SRS effect.

Figure 5(a) shows the time traces at different output powers. We can obtain that the fluctuation of laser intensity will be stronger with increase of the output power. By applying Fourier analysis on the time traces, we obtained the frequency distribution of the beam fluctuation, as shown in Fig. 5(b). At the power of 1760 W, the frequency components of the Fourier spectrum distributed mainly from 1 kHz to 2.2 kHz, which means that the sign of the MI in high-power fiber laser appeared. It can be seen that the frequency components tend to drift to higher frequency gradually with the growth of the output power. The MI in the high-power fiber laser refers to the rapid fluctuations when the output power exceeds the threshold. Accordingly, the beam quality deterioration at high output power is related to the MI effect.[42]

The variation of beam quality factor with the laser intensity in the fiber core area is depicted in Fig. 6. The M^2 value of fiber laser is usually measured by the beam quality analyzer using the second moment method.[43] While the laser intensity in fiber core region changes gradually from 213 MW/cm^2 to 254 MW/cm^2 corresponding to the output power from 1510 W to 1800 W, the beam quality deteriorated slightly from 1.21 to 1.49. When the output power reaches 1760 W, the M^2 factor suddenly rises from 1.21 to 1.49 which indicates the onset of MI. Beyond that power level, the beam quality keeps degrading. However, the M^2 factor begins to grow due to the newly generated spectral components and MI effect as the intensity exceeds 393 MW/cm^2 (2784 W). Finally, the beam quality tends to stabilize with the M^2 of 2.80 and the far-field beam profile is exhibited in the inset of the figure. It can be seen from the far-field beam profile that the propagation modes in the optical fiber contained the fundamental mode (LP$_{01}$ mode) and the secondary mode (LP$_{11}$ mode). Therefore, the MI and FWM effects on high output power fiber laser have an important influence on beam quality.

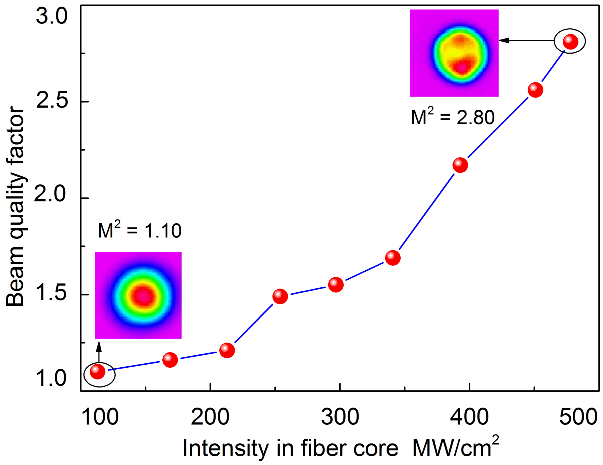

Fig. 6. Variation of the beam quality factor with the intensity in fiber core.

5. Conclusion

In this paper, we have researched and demonstrated successfully the FWM effect on high-power all-fiber laser. The detail theoretical description of phase matching was presented and we found that the frequency shift is about 5.16 THz when we choose LP_{11} mode as the Stokes light. Moreover, the FWM products, corresponding to the wavelengths of 1100 nm and 1060 nm, were also observed in high-power all-fiber laser system as the intensity in fiber core increased to 394 MW/cm^2. In addition, the newly generated spectral components and MI effect in high-power fiber laser could deteriorate the beam quality. The M^2 factor measured at the maximal intensity of 478 MW/cm^2 is 2.80.

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